KRISHNA CLASSES

WORKSHEET

SUBJECT- MATHS

CHAPTER- 3

NAME OF STUDENT –

ISSUE DATE –

Chapter-3

1. If a pair of linear equations is consistent, then the lines will be
   (a) Always coincident
   (b) Parallel
   (c) Always intersecting
   (d) Intersecting or coincident
2. The pair of equations $x + 2y - 5 = 0 \; and \; -4x - 8y + 20 = 0$ have
   (a) Unique solution
   (b) Exactly two solutions
   (c) Infinitely many solutions
   (d) No solutions
3. If the lines given by $3x + 2ky = 2 \; and \; 2x + 5y + 1 = 0 \; are$ parallel, then the value of k is:
   (a) 4/15
   (b) 15/4
   (c) 5/4
   (d) 3/2
4. The pair of equations $x = a \; and \; y = b \; graphically \; represents \; lines \; which \; are$:
   (a) Parallel
   (b) Coincident
   (c) Intersecting at (a, b)
   (d) Intersecting at (b, a)
5. The sum of the digits of a two digit number is 9. If 27 is added to it, The digits get reversed. The number is
   (a) 25
   (b) 72
   (c) 63
   (d) 36
6. The value of k for which the system of equations
   $x + 2y = 3 \; and \; 5x + ky + 7 = 0$ has no solutions
   (a) 10
   (b) 6
   (c) 3
   (d) 5
7. The area of triangle formed by the lines $y = x, \; x = 6 \; and \; y = 0 \; is$:
   (a) 36 sq. unit
   (b) 18 sq. unit

(c) 9 sq. unit
(d) 72 sq. unit

8. Aruna has only 1Rs. And 2 Rs. coins with her. If the total number of coins she has is 50 and the amount of money with her is 75Rs., then the number of 1Rs. And 2 Rs. coins are, respectively.
   (a) 35 and 15
   (b) 15 and 35
   (c) 25 and 25
   (d) 30 and 20
9. The father's age is six times his son's age. Four years hence, the age of the father will be four times his son's age. The present age of the son and the father are:
   (a) 4 and 24
   (b) 5 and 30
   (c) 6 and 36
   (d) 3 and 18
10. If $2x + 3y = 11$ *and* $2x - 4y =- 24$ then the value of m for which $y = mx + 3$ is:
   (a) -1
   (b) 1
   (c) 5
   (d) -2
11. Aftab tells his daughter, "Seven years ago, I was seven times as old as you were then. Also, three years from now, I shall be three times as old as you will be." (Isn't this interesting?) Represent this situation algebraically and graphically.
12. The cost of 2 kg of apples and 1kg of grapes on a day was found to be Rs.160. After a month, the cost of 4 kg of apples and 2 kg of grapes is Rs.300. Represent the situation algebraically and geometrically.

13. Which of the following pairs of linear equations are consistent or inconsistent? If consistent, obtain the solution graphically:

(i) $x + y = 5, 2x + 2y = 10$

(ii) $x - y = 8, 3x - 3y = 16$

(iii) $2x + y - 6 = 0, 4x - 2y - 4 = 0$

(iv) $2x - 2y - 2 = 0, 4x - 4y - 5 = 0$

14. Solve the following pair of linear equations by the substitution method
   i. $x + y = 14,\ x - y = 4$

ii. $s - t = 3$, $\frac{s}{3} + \frac{t}{2} = 6$

iii.

$\frac{b}{a}x + \frac{a}{b}y = a^2 + b^2$, $x + y = 2ab$

15. The larger of two supplementary angles exceeds the smaller by 18 degrees. Find them.

16. A fraction becomes 9/11 , if 2 is added to both the numerator and the denominator. If, 3 is added to both the numerator and the denominator it becomes 5/6. Find the fraction.

17. Five years hence, the age of Jacob will be three times that of his son. Five years ago, Jacob's age was seven times that of his son. What are their present ages?

18. Solve the following pair of linear equations by the elimination method

(i) $x + y = 5$ and $2x - 3y = 4$

(ii) $3x + 4y = 10$ and $2x - 2y = 2$

(iii) $141x + 93y = 189$;
$93x + 141y = 45$

19. The sum of the digits of a two-digit number is 9. Also, nine times this number is twice the number obtained by reversing the order of the digits. Find the number.

20. Meena went to a bank to withdraw Rs.2000. She asked the cashier to give her Rs.50 and Rs.100 notes only. Meena got 25 notes in all. Find how many notes of Rs.50 and Rs.100 she received.

21. A lending library has a fixed charge for the first three days and an additional charge for each day thereafter. Saritha paid Rs.27 for a book kept for seven days, while Susy paid Rs.21 for the book she kept for five days. Find the fixed charge and the charge for each extra day.

22. The age of the father is twice the sum of the ages of his 2 children. After 20 years, his age will be equal to the sum of the ages of his children. Find the age of the father

23. A boat goes 30 km upstream and 44 km downstream in 10 hours. In 13 hours, it can go 40 km upstream and 55 km downstream. Determine the speed of the stream and that of the boat in still water.

24. A two-digit number is such that the product of its digits is 18. When 63 is subtracted from the number, the digits interchange their places. Find the number.

25. A fraction becomes 1/3 when 1 is subtracted from the numerator and it becomes 1/4 when 8 is added to its denominator. Find the fraction.

26. In a test, 3 marks are given for every correct answer and 1 mark is deducted for every wrong answer. Yash scored 40 marks. Had 4 marks been awarded for each correct answer and 2 marks been deducted for each incorrect answer, Yash would have scored 50 marks. How many questions were there in the test?

27. The area of a rectangle gets reduced by 9 square units, if its length is reduced by 5 units and breadth is increased by 3 units. If we increase the length by 3 units and the breadth by 2 units, the area increases by 67 square units. Find the dimensions of the rectangle.

28. Places A and B are 100 km apart on a highway. One car starts from A and another from B at the same time. If the cars travel in the same direction at different speeds, they meet in 5 hours. If they travel towards each other, they meet in 1 hour. What are the speeds of the two cars?

29. 2 women and 5 men can together finish an embroidery work in 4 days, while 3 women and 6 men can finish it in 3 days. Find the time taken by 1 woman alone to finish the work, and also that taken by 1 man alone.

30. Roohi travels 300 km to her home partly by train and partly by bus. She takes 4 hours if she travels 60 km by train and the remaining by bus. If she travels 100 km by train and the remaining by bus, she takes 10 minutes longer. Find the speed of the train and the bus separately.